

Physics of the Atmosphere

Student sheet

Name: _____

Date: _____

Safety: On the stairs, walk, do not run; hold the box with both hands and read the screen only standing on a landing.

1. What I see on the screen

The screen shows **Temperature**, **Humidity** and **Pressure** (hPa, two decimal places) and, computed from them, **Dew point** and **Heat Index** (the felt temperature). **Heat Index** needs above 27°C and above 40% humidity. If either is missing, the app shows the air temperature itself there — **do not record it as a result**.

2. What I do on my own

First I predict. Climbing the stairs, pressure will: ☐ increase ☐ decrease ☐ stay the same, by _____ hPa per storey.

because _____

Building: _____ **One storey:** _____ m **Forecast QNH:** _____ hPa

Baseline measurement. $T =$ _____ °C $RH =$ _____ % $P =$ _____ hPa

$z = 44330 [1 - (P/P_0)^{0.1903}]$: with $P_0 = 1013.25$: _____ m; with $P_0 = \text{QNH}$: _____ m true:

_____ m

$\gamma = \ln \frac{RH}{100} + \frac{17.625 T}{243.04 + T} =$ _____ $T_d = \frac{243.04 \gamma}{17.625 - \gamma} =$ _____ °C $T - T_d =$ _____ °C

The traverse. 2–3 min per landing. $\ln P$ to 5 decimal places: $1008.40 \Rightarrow 6.91612$.

Floor	z (m)	T (°C)	P (hPa)	$\ln P$ (5 d.p.)
Ground floor (mean)	0			
Floor 1				
Floor 2				
Top floor (mean)				
Ground floor (re-	0			
turn)				

The check. Three readings at the ground floor and three at the top, 30 s apart; enter the mean.

Ground floor: _____ spread (max – min) = _____ hPa

Top floor: _____ spread = _____ hPa

Difference between levels = _____ hPa. A larger spread means **not detectable**, not that it does not exist: ☐ detectable ☐ not detectable

The return to the ground floor is not optional: a different reading means the weather changed and the gradient cannot be trusted. Difference _____ hPa. Trust it? ☐ YES ☐ NO

The graph. $\ln P$ vertical, z horizontal. **The vertical axis does not start at zero.**

top: _____

bottom: _____ Height z above the ground floor (m), from 0 to _____ — ten divisions

Gradient, from two points *on the line*: $z_1 = \text{_____ m}$, $\ln P_1 = \text{_____}$ $z_2 = \text{_____ m}$, $\ln P_2 = \text{_____}$
 $m = (\ln P_2 - \ln P_1)/(z_2 - z_1) = \text{_____ m}^{-1}$ $H = -1/m = \text{_____ m}$ against 8400 m: _____ %

3. The conclusion I write

I predicted that, and the data show that _____

The difference is explained by _____

The gradient of my line is the scale height of the atmosphere, so _____

Exit ticket

One thing that surprised you today, and one thing you would do differently if you repeated the measurement.
